

Super-Eddington Little Blue Dots Over-Reionize Helium

arXiv:2608.28745 — Super-Eddington Little Blue Dots May Reionize Helium Too Early

First author / year / source
arXiv:2608.28745 doi:10.48550/arXiv.2608.28745



JWST's faint AGN may complete helium reionization too early unless only a small fraction are super-Eddington.

Background

JWST found many faint AGN at $z > 4$, the Little Blue Dots, possibly super-Eddington with weak X-rays. Their ionizing output could reshape helium reionization timing.

Do observed Little Blue Dots reionize helium too early, conflicting with helium reionization ending near $z \approx 3$?

What this paper is selling

The paper tests whether super-Eddington LBDs dominate helium reionization. It finds that if they do, only $\lesssim 10\%$ can be super-Eddington, otherwise helium reionizes too early.

Key results to remember

- Helium reionization ending at $z \approx 3$ limits super-Eddington LBDs to $\lesssim 10\%$.
- Majority super-Eddington requires escape fractions $\lesssim 1.5\%$ or lower abundance.
- Constraints depend on uncertain extreme-UV spectra of super-Eddington black holes.

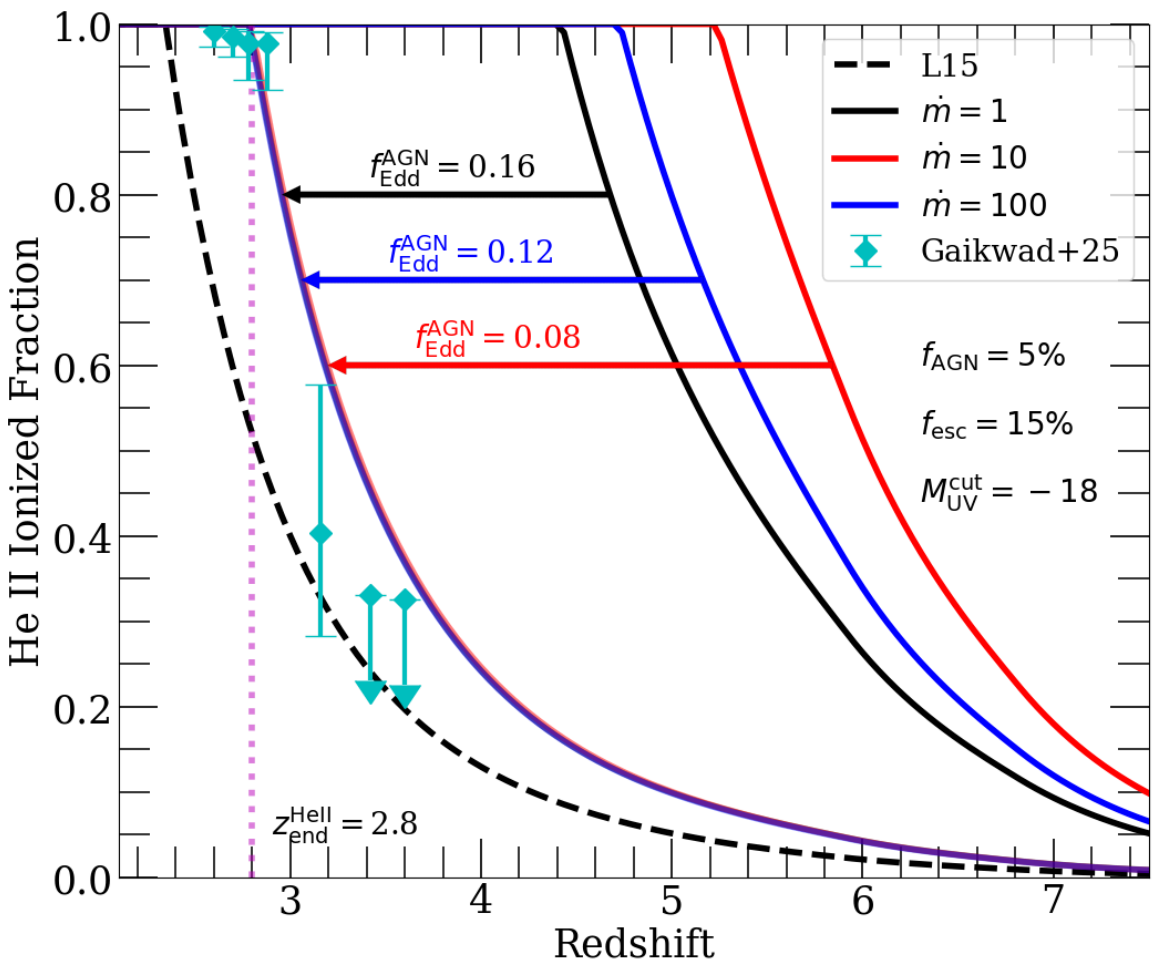


Fig. 3. Helium reionization histories show that super-Eddington models complete reionization earlier than allowed by observations.

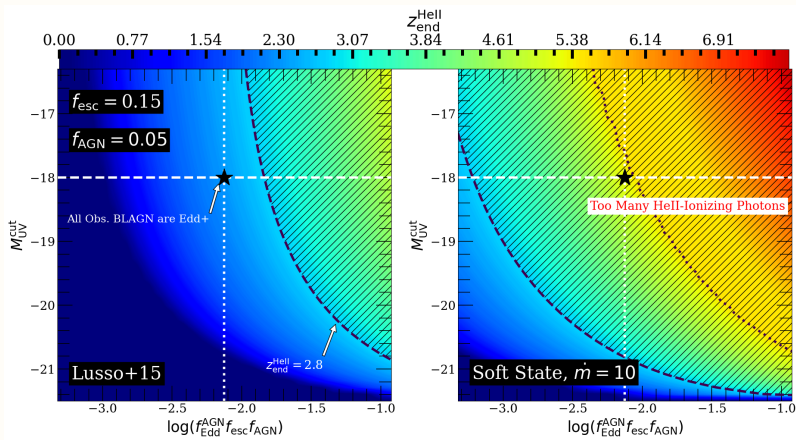


Fig. 4. Excluded parameter space: hatched regions produce helium reionization ending at $z \geq 2.8$, disfavored by current constraints.